# Sovereign Trade Copilot

## A traceable first reply for human approval

Track 2B | Online Hack Apertus | Team: Sovereign Trade Copilot | 5 October 2026 Report and new synthetic data: CC-BY-4.0

## 1. Purpose

A professional-looking reply can invent company capabilities or commercial terms. This prototype prepares a useful first reply for a person, rather than certifying it safe to send. It processes one new inquiry, with no automatic sending, quotation execution, payment, production, CRM or memory expansion.

## 2. Final architecture and Apertus role

Original inquiry + source-backed company conditions + declared source state + company-neutral Trade Judgment guidance -> short business brief -> explicit context confirmation when needed -> one Apertus draft -> small phrase-located warnings -> human editing and explicit approval.

The brief preserves original source statements and applicable scope. An optional operator focus chooses business direction; it cannot authorize absent company claims. Guidance prioritizes reliable current answers, up to three decision-changing questions, No Re-ask, reversible advances and conditional rough costing. Unknown is not automatically a blocker. Performance targets are not company achievements. Unread material must not become reviewed or missing.

This is not full typed extraction, deterministic judgment, AUTO message planning, slot rendering, generation retry or fresh-audit authority. Apertus writes plain customer-facing text from the complete inquiry and compact brief. Requested and returned model: swiss-ai/Apertus-v1.5-70B through CSCS; temperature 0.1; maximum 1,100 output tokens; one transport attempt.

Company conditions need statement, source, scope and status. Explicit operator authorization is required for confirmed company claims; public/working references remain separate. Mechanical checks validate input shape, source declaration, authorization, review and draft-version identity. Natural-language warnings are advisory, incomplete and phrase-located; no automated safe-to-send label exists.

## 3. Synthetic data and first-output evaluation

Three original synthetic cases and invented company conditions cover known terms, quotation-tier purpose/branded sourcing, and one prototype with unread material. No private customer record, company seed or string-redacted real inquiry is released.

Inputs, business gold and complete prompts were frozen before model calls. Each case received one first draft: no retries or best-of selection. The development agent performed semantic review; this is not independent blind human review. Business focus was curated. Automated UI approvals are not evidence of human sendability, review speed or daily usefulness.

| Case | Raw grade | Key result |
|---|---|---|
| Hero | B: minor edit | Reliable terms/conditions answered; two useful request groups; explicit assessment-to-costing next move missing |
| Quote tiers | E: commercial risk | General MOQ became per style; parallel conditional rough costing omitted |
| One prototype | E: commercial risk | Sample-fee inclusions invented; unread source stayed pending and known one-prototype quantity was not re-asked |

E describes commercial error in the raw text, even when its correction is short. The two E outputs are not counted as safety successes after editing. Three real calls used 3,673 tokens: 3,278 input and 395 output. Summed client latency was 10.340 seconds; median was 3.290 seconds. These are generation measurements, not reviewer productivity or matched before/after performance.

## 4. Hero: measured first draft and required edit

The actual raw draft is preserved in data/light-demo/first-drafts.json. It answers custom-category fit, the general 240-piece baseline, 18 business days after fee receipt/material availability, and 70-95 calendar days after deposit and sample approval subject to material/capacity checks. It asks for priority design/material specs and first-production quantity/style-color allocation.

Required operator addition: "Once we review the priority design and material route, we can prepare a sample-cost basis and conditional preliminary costing." A clarity edit may narrow the vague minimums-review clause to allocation review. This edit was prepared by the development agent; no human has approved it.

## 5. Failure details and previous negative evidence

Quote-tier raw output invented "240 pcs per style" although only a general baseline was authorized. The original warning caught "per style". Review must remove that basis and restore conditional comparison-tier costing, subject to material-route verification and lower-tier internal approval. The draft did not guarantee branded sourcing or convert tiers into an order.

Prototype raw output invented fee coverage of pattern making, grading and initial sewing, and unnecessarily requested permission to begin already-requested work. Review must remove those fee inclusions and retain parallel conditional costing alongside source/material review. Checking fabrics against targets is verification intent, not a confirmed performance result.

A generic fee inclusion/refund/credit advisory flag was added after generation. It now locates "fee to cover" on unchanged playback. The original warning record correctly retains that this was missed. Inputs, prompts, expectations and raw drafts were unchanged. Next-move omission still requires human review. Three cases cannot establish warning precision or recall.

An earlier owner-supplied ChatGPT blind review of three scenarios/nine drafts gave: full v2/P5.2 E / NONE / D; concise Apertus direct E / E / E; native Codex A / A / A. The preferred native session contained prior human calibration and gold discussion. It was not an isolated causal model comparison. Its texts were preferred, but model superiority and strong-model behavior under the full Harness were untested.

Full-chain traces showed product/quantity ownership errors, required-answer loss and invalid reply JSON. Direct drafts fabricated capabilities and commercial scope. These findings motivated reducing automatic intermediate authority while preserving source boundaries and human approval. Only aggregate non-identifying findings are released. This new evaluation is not a matched improvement claim.

## 6. Remaining limitations

Curated business focus is not validated automatic planning. Source authenticity is operator-asserted, not independently verified. The UI does not parse binary attachments. Three English cases do not establish multilingual reliability, generalization, commercial safety or daily productivity. Capability claims and omissions can escape warnings. No timed human study or final independent human sendability review has been completed.

## 7. Deployment, privacy and reproducibility

Target: on-premise application using an operator-controlled OpenAI-compatible Apertus endpoint. Runtime is Python standard library only, with a self-contained page and no browser CDN/analytics. The final Docker image includes only Light Harness files, the UI and the new synthetic JSON. Old chains and private seeds are excluded. Local weights, GPU provisioning and an on-premise weights server are not included or tested. Measured inference used the authorized CSCS endpoint. Air-gapped inference would require an actual local Apertus server; recorded playback is not local LLM inference. See the official model-card serving examples.

Keys enter through LLM_NAME, LLM_BASE_URL and LLM_API_KEY runtime variables, never source/image/log files. Drafts and approvals occupy bounded, ephemeral memory; restart clears them. The Makefile publishes loopback access. Same-origin writes and no request-body logging reduce exposure. This is a local prototype, not an authenticated multi-user production service.

Run make run at repository root or in track_2b; open http://localhost:8000/ui. Set LLM_* before startup for new generation. Without an endpoint, exact unchanged synthetic cases replay their measured first drafts. Altered input requests a live endpoint. Run python src/selftest_light.py for source/authorization, no-retry, cold-start, original preservation and approval-version boundary checks.

An actual clean Git clone passed Docker make run, health/UI, three exact recorded POST /process cases, changed-input rejection and scripted version approval. A separate Docker-contained live Apertus call passed in one attempt: 1,245 tokens, 5,208 ms, returned v1.5-70B. This proves integration, not better business quality; it is not a selected evaluation retry. Six grouped runtime boundary checks passed. Browser editing and approval were exercised; privacy checks found no known key in application files or logs. See docs/RELEASE-VERIFICATION.md for actual scope.

## 8. Demo and release interpretation

The video demonstrates actual recorded first-output review, explicit manual correction and scripted approval, not a timed human study. No email is sent. The local artifact is a 90-second, 1080p H.264 video with explanatory on-screen copy. Publication and the demo-video URL are pending this round's no-push boundary. Freeze can cover a reproducible review demo, not an autonomous First Reply service. The boundary errors remain visible; architecture expansion is stopped.

## References

Official competition: https://hackapertus.ch/online-hack Official model card: https://huggingface.co/swiss-ai/Apertus-v1.5-70B Reproducible evidence: data/light-demo/cases.json, freeze.json, first-drafts.json Business review: docs/LIGHT-DEMO-EVALUATION.md